

Functions

Formally: A function f from a set D to a set E is a correspondence that assigns to each element x of D exactly one element y of E ($f: D \rightarrow E$)

Informally: A function is a rule that assigns to every valid input a unique output

The set D or the collection of "valid inputs" form the domain of the function

the collection of all possible "outputs" forms the range of the function.

e.g. function that assigns to each human being their birth-date

domain = set of all "valid inputs"
= {any human}

range = {a collection of ~~birth~~ all dates from time of 1st human till the present day}

What is the domain and range of $f(x) = x^2$?

set of all
"valid inputs"
set of all
outputs

domain is (any real number works) $\mathbb{R}$
OR $(-\infty, \infty)$

range is $[0, \infty)$ (anything ≥ 0)

e.g. What is the domain of $\frac{\sqrt{4+x}}{1-x}$?

"valid input" here makes both
the numerator and denominator
well-defined & we need
denominator to be $\neq 0$

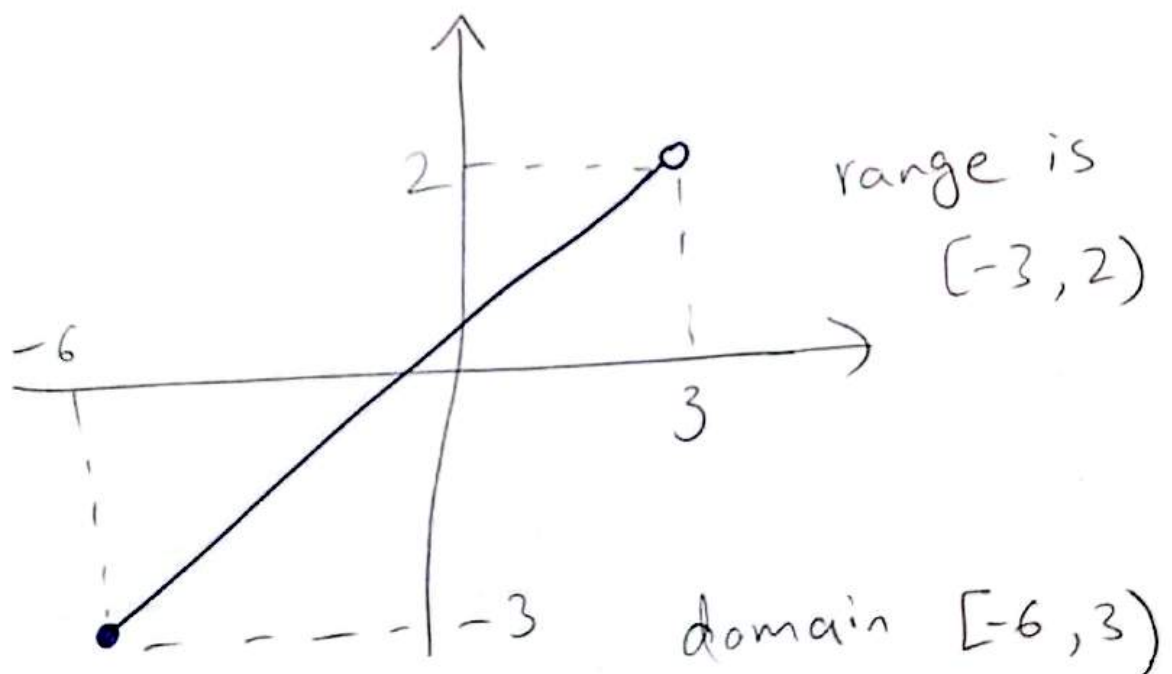
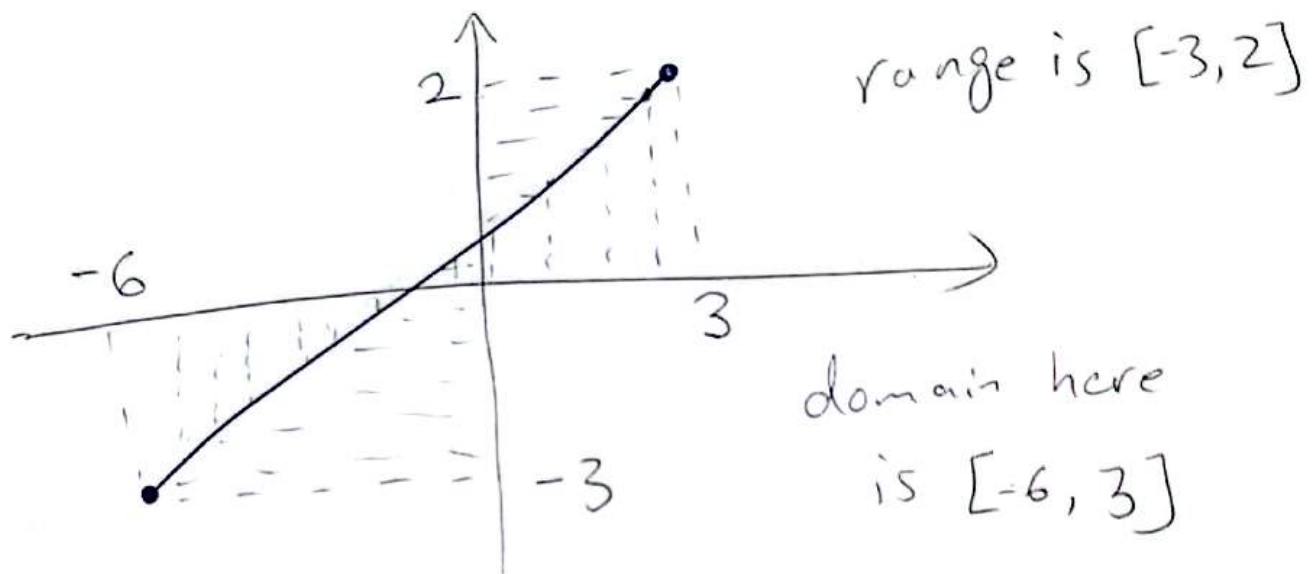
$$x \neq 1 \quad \& \quad x \geq -4$$

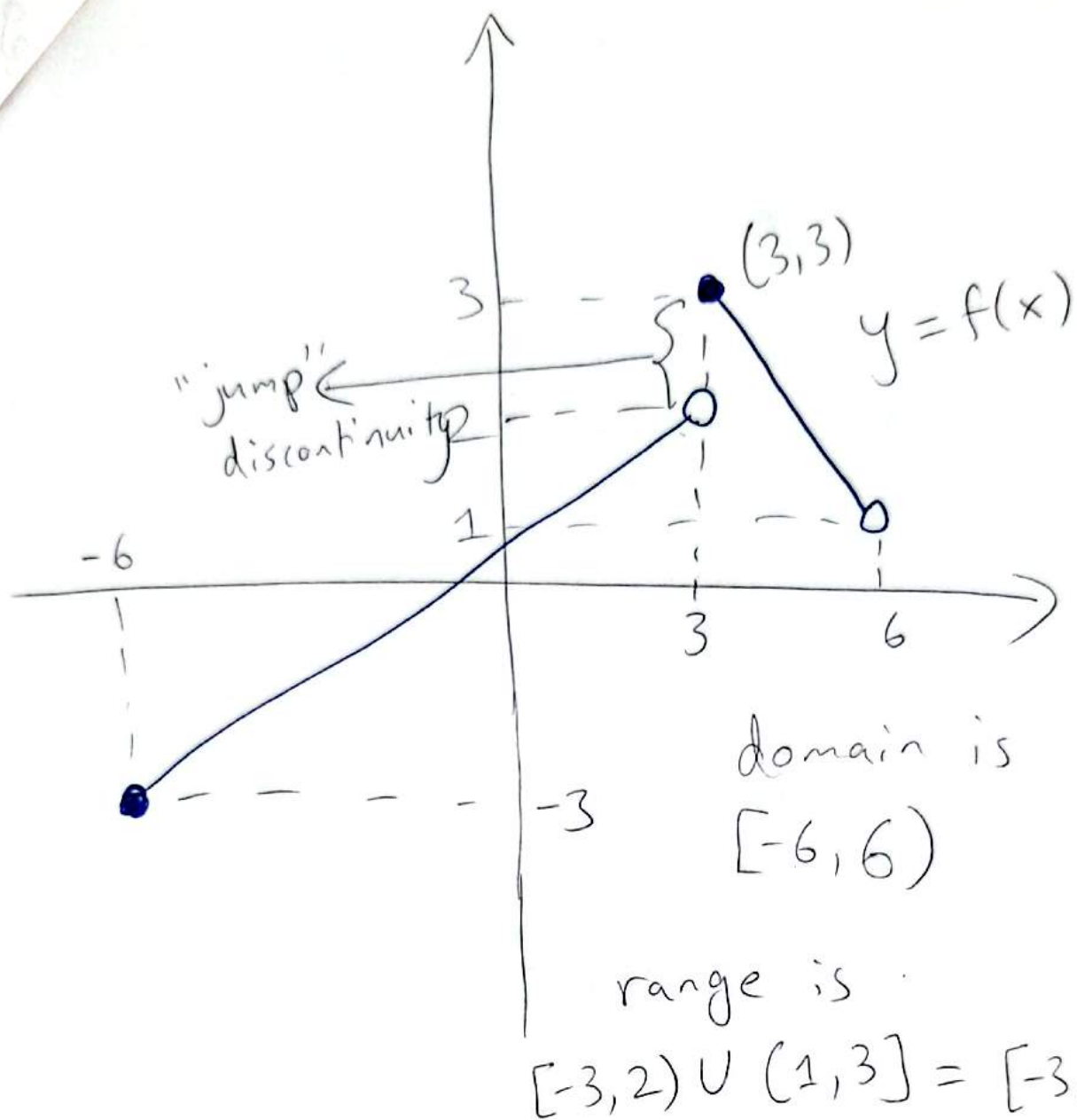
$$(-\infty, -4) \cup [-4, 1) \cup (1, \infty)$$

domain
in interval
notation

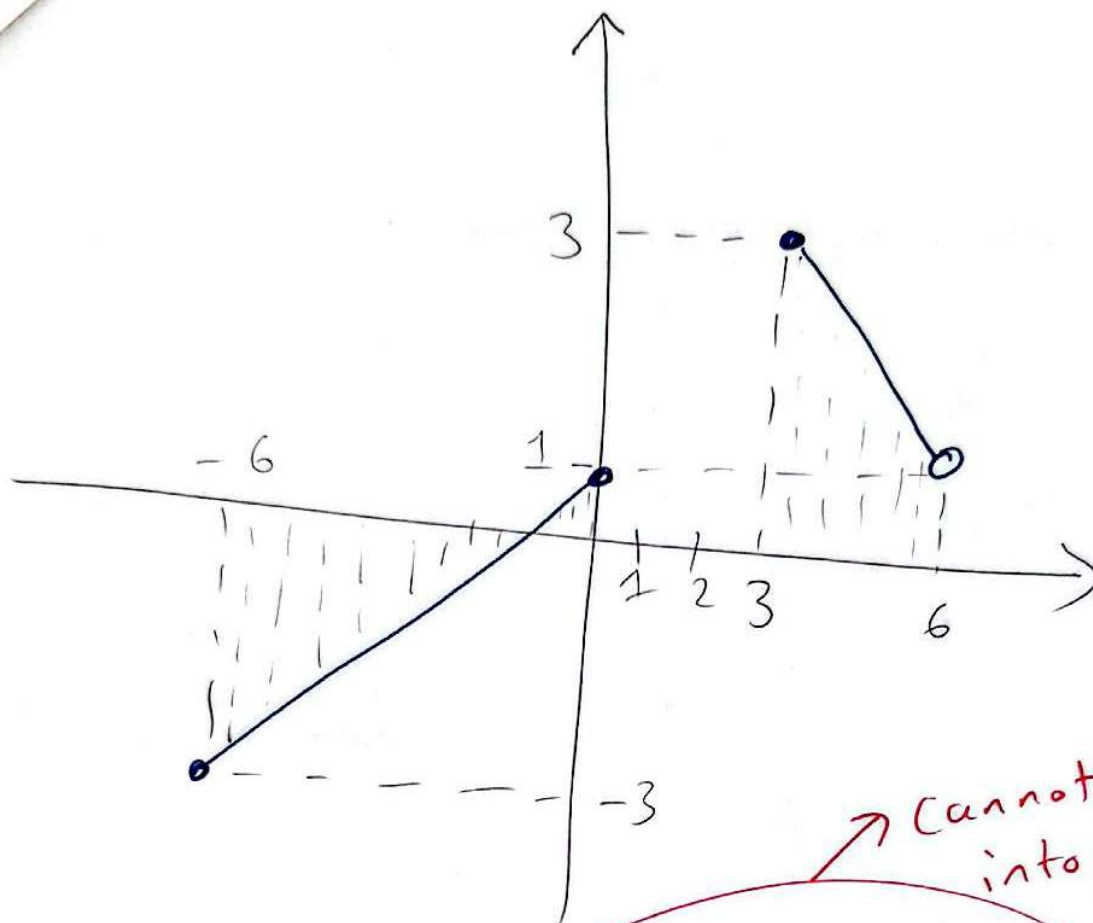
Reading Domains and Ranges

Graphically





When $y = f(x)$, the domain is the horizontal spread of the graph & the range is the vertical spread of the graph



$$\text{domain} = [-6, 0] \cup [3, 6)$$

we're "missing"
whole interval

$\neq (0, 3)$ from domain

$$\text{range} = [-3, 1] \cup (1, 3]$$

$$= [-3, 3]$$

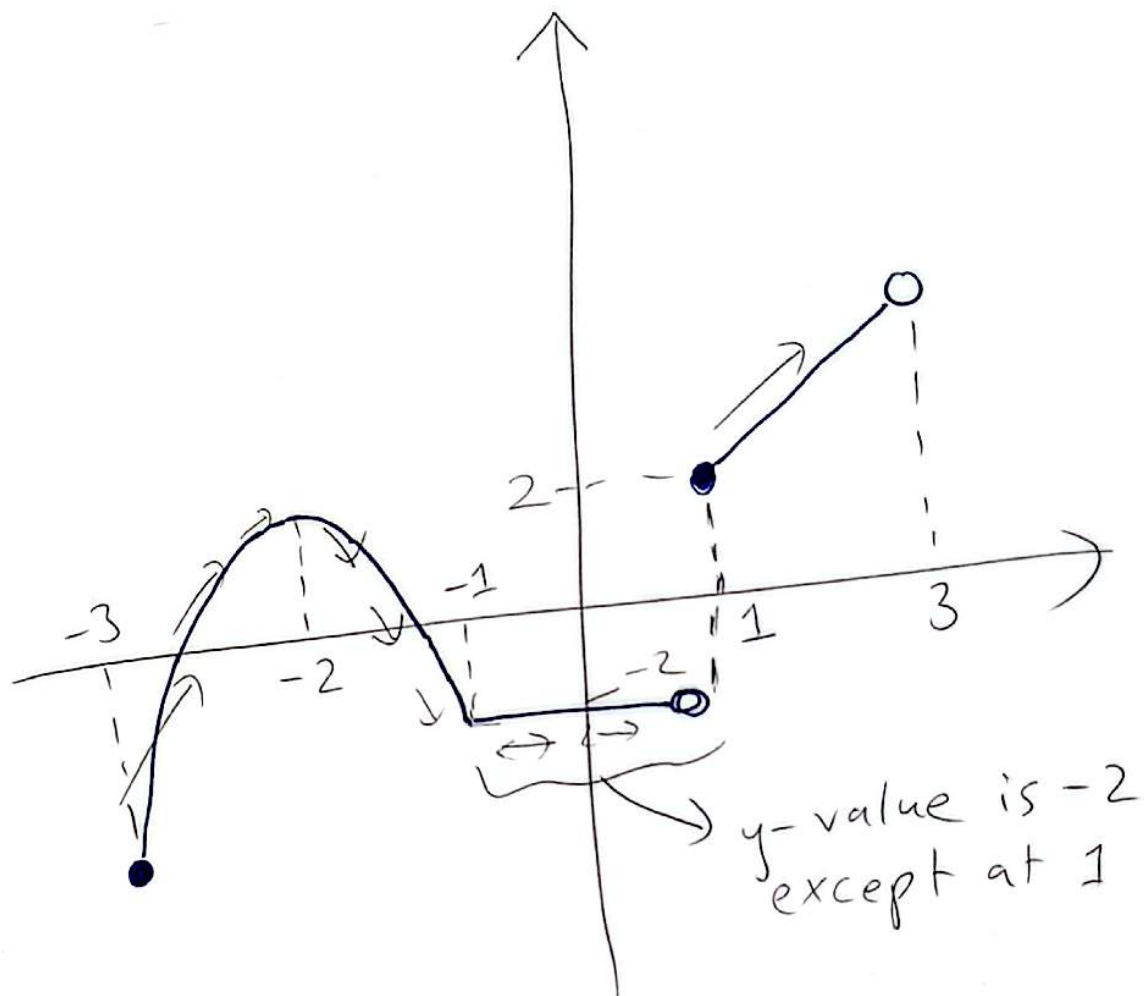
bigger inputs
= bigger outputs

Terminology	Definition	Graphical interpretation
f is increasing on an interval I	$f(x_1) < f(x_2)$ whenever $x_1 < x_2$	
f is decreasing on an interval I	$f(x_1) > f(x_2)$ whenever $x_1 < x_2$	
f is constant on an interval I	$f(x_1) = f(x_2)$ for every x_1 and x_2	

bigger inputs
= smaller outputs

flat graph
("horizontal")

Leading Increasing/Decreasing Behavior Graphically



domain = $[-3, 3)$

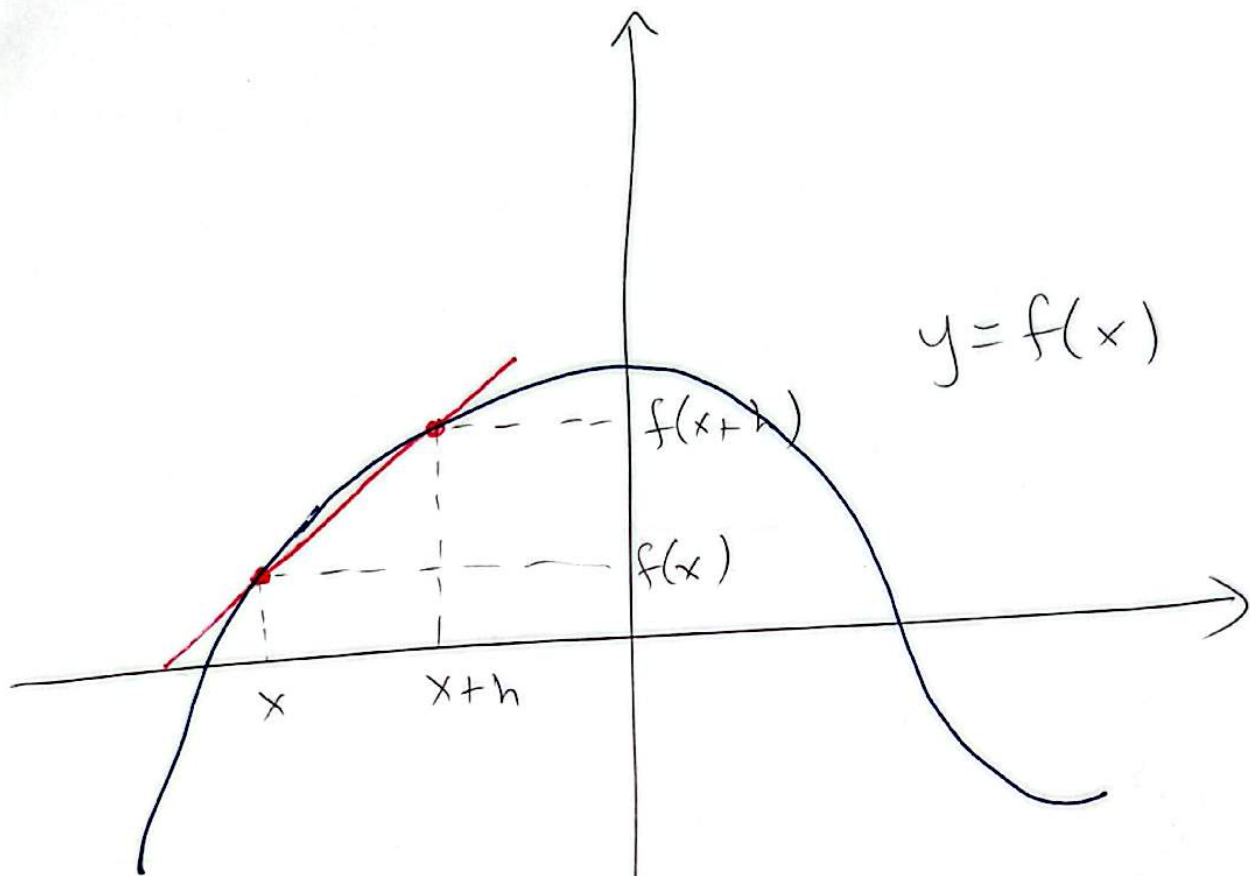
Where is this function increasing?

$[-3, -2] \cup [1, 3)$

It's decreasing $[-2, -1]$
And it's constant $[-1, 1]$

incr. on $[-3, -2]$
and $[1, 3)$ separately

Difference Quotients



Secant Line

has slope $\frac{f(x+h) - f(x)}{(x+h) - x}$

$$= \boxed{\frac{f(x+h) - f(x)}{h}}$$

Difference Quotient

Find the difference quotient of $f(x) = x^2 + 6x - 4$ and simplify

$$\frac{f(x+h) - f(x)}{h}$$

$$= \frac{f(x+h) - [x^2 + 6x - 4]}{h}$$

$$= \frac{[(x+h)^2 + 6(x+h) - 4] - [x^2 + 6x - 4]}{h}$$

binom. squared

$$= \frac{\cancel{x^2} + 2xh + h^2 + \cancel{6x} + 6h - \cancel{4} - \cancel{x^2} - \cancel{6x} + \cancel{4}}{h}$$

$$= \frac{2xh + h^2 + 6h}{h} = \frac{(2x + 6 + h)h}{h}$$

$$= \boxed{2x + 6 + h}$$